

Ranking Is Not Threshold Portability: Validating IFRS 16 Covenant Screening in a Synthetic Benchmark

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Abstract

Covenant screening requires separate validation of a mechanically consistent financial engine, a score that ranks adverse outcomes, and a decision threshold that transfers across borrower structures. We examine these layers in a synthetic IFRS 16 benchmark of 200 scenarios across five archetypes, including 20 broad financial failures. Archived independent checks and runtime invariants support the engine mechanics on the tested paths. The fixed screening score achieves ROC-AUC 0.947 and average precision 0.781. Leave-one-template-out thresholds selected only on the training archetypes raise recall from 3/20 (15.0%) under the fixed 0.5 comparator to 15/20 (75.0%), but generate 38 false positives. This trade-off is sharply heterogeneous: Template 004 contributes 37 of 38 false positives, and its balanced accuracy is 0.513 under the transferred threshold versus 0.636 at 0.5. The score is not a calibrated probability of default, and sparse failures and overlapping training sets constrain inference beyond this synthetic realization. Strong ranking therefore did not imply threshold portability across synthetic borrower archetypes. Engine correctness, ranking performance, and decision usefulness require separate validation.

Keywords: IFRS 16; covenant screening; synthetic benchmark; threshold transfer; model validation.

1 Introduction

IFRS 16 brings most lessee lease commitments onto the balance sheet, subject to the standard’s exemptions ([IFRS Foundation, 2016](#)). In covenant screening, a lease-inclusive calculation and a stipulated rent-adjusted alternative need not represent the same contractual definition. The accounting convention therefore needs to be stated before interpreting a leverage or coverage test. Neither calculation, by itself, establishes the terms of an actual loan.

The distinction between a covenant condition and a missed payment also matters for the interpretation of a screening outcome. The creditor-control perspective examined by [Nini et al. \(2009\)](#) motivates separating covenant conditions from debt-service events.

This paper asks how much a successful validation result establishes about the next decision layer. A financial simulation may reconcile its cash and debt mechanics, while a screening score ranks its adverse outcomes well. Those findings do not establish that a threshold selected on one collection of structures will work on another. The question is not exhausted by the discrimination of the score: the usefulness of a threshold requires its own evaluation.

The study is deliberately synthetic. Its contribution is a bounded demonstration of the distinction between mechanical correctness, ranking performance and threshold portability across stipulated borrower archetypes. It does not estimate borrower-population performance or claim that the observed transfer difficulty occurs with the same frequency outside this system. The

central thesis is a possibility demonstrated within the benchmark: strong ranking can coexist with an unstable decision rule.

The organization follows that distinction. We first establish what the mechanical validation supports, then separate the fixed score from the procedure used to select a threshold. Aggregate results are followed by the held-out template evidence, so that a favorable aggregate does not substitute for the examination of transfer. The limitations return to the scope of each validation layer.

2 Research question and contribution

The research question is whether a mechanically validated screening workflow with strong synthetic discrimination also supports a portable threshold across structurally different synthetic archetypes. The answer in this benchmark is qualified: ranking remains strong, but the transferred thresholds do not provide a stable rule across all the archetypes. This is the paper’s primary conceptual result, rather than a claim of universal failure or universal success for screening models.

Figure 1 distinguishes the three layers. Engine correctness concerns the consistency of the tested financial mechanics. Ranking performance concerns the ordering of adverse outcomes by the screening score. Decision usefulness concerns the behavior of the resulting threshold rule. Evidence at one layer must not be read as evidence that the next layer has passed. This distinction determines both the presentation of the results and the limits placed on their interpretation.

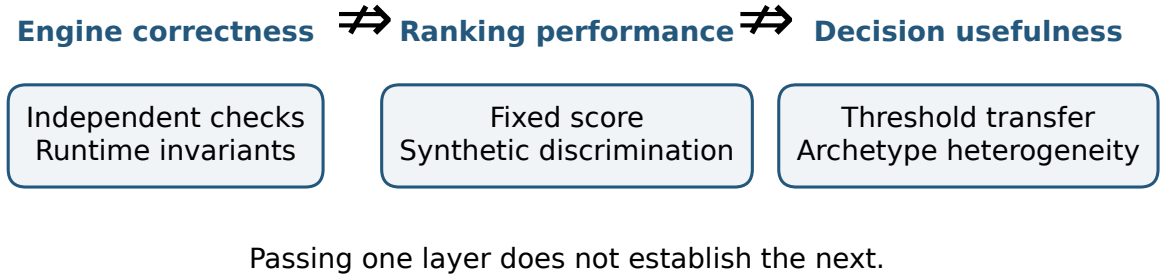


Figure 1: Three distinct validation layers. Passing one layer does not establish the next. This is a conceptual synthesis of the frozen validation evidence, not a causal diagram or a claim about external borrower performance.

3 Related literature and validation perspective

Debt covenants turn accounting measurements into contractual control points. Large-sample evidence on accounting-based covenants shows that firms cluster close to covenant thresholds and that apparent technical violations are economically meaningful rather than rare accounting curiosities (Dichev and Skinner, 2002). Covenant violations can transfer control rights to creditors and alter firm investment policy, which gives the measured boundary consequences beyond the label attached to a screen (Nini et al., 2009). Contract design also distinguishes capital covenants, which constrain balance-sheet positions, from performance covenants that act as trip wires when creditor claims become vulnerable (Christensen and Nikolaev, 2012). These findings do not determine the contractual definition for a synthetic benchmark. They instead explain why the accounting convention, breach rule and tested financial mechanics must be explicit. IFRS 16 changes the recognition of most lessee obligations, subject to exemptions (IFRS Foundation, 2016); whether

a credit agreement incorporates, freezes or adjusts that accounting treatment remains a separate contractual question. In the framework used here, this concern belongs first to engine correctness: a screen cannot be interpreted until the quantities entering it are calculated consistently with the stated convention.

Ranking quality answers a different question. ROC analysis evaluates how a score orders positives and negatives as its threshold varies, and it is particularly useful when class frequencies or error costs make raw accuracy hard to interpret (Fawcett, 2006). A high ROC-AUC can therefore support a discrimination statement without identifying a useful operating threshold. Hand (2009) goes further by showing that an AUC summary embeds assumptions about the relative costs of errors and can be misleading for comparison when those assumptions do not match the decision problem. The broader prediction-model literature likewise treats discrimination, calibration and decision-analytic value as separate dimensions of performance (Steyerberg et al., 2010). That separation is central here. The screening score is a monotone ordering device derived from analytic headroom; it has not been calibrated as an event probability. Its AUC and average precision describe ranking within the frozen benchmark, while the confusion counts describe what happens after a particular decision rule is applied.

Threshold choice adds an objective that ranking metrics do not supply. Decision-curve analysis formalizes this point by evaluating predictions through the relative consequences of false-positive and false-negative actions over candidate thresholds (Vickers and Elkin, 2006). That literature is developed for clinical decisions, but its narrow methodological lesson carries over: a threshold is useful only in relation to an action and the costs attached to the two kinds of error. Balanced accuracy gives equal weight to sensitivity and specificity and is a transparent selection objective for this experiment. It is not an estimate of economic utility. A transferred rule can improve aggregate balanced accuracy while imposing an unacceptable burden in one borrower structure, so threshold evaluation must retain both confusion counts and subgroup denominators.

Transport adds a third distinction. Dataset shift refers to changes in the joint distribution of inputs and outcomes between training and use, with covariate shift as the special case in which the input distribution changes (Quiñonero-Candela et al., 2008). The five archetypes in this study create a controlled form of structural heterogeneity rather than a sample from an identified borrower population. Leave-one-template-out evaluation asks whether a threshold chosen on four stipulated structures behaves similarly on the fifth. It is a stronger check of within-benchmark transfer than reapplying a threshold to its fitting rows, but it does not identify performance under an external population distribution. The overlapping training sets and sparse positive counts further limit any population-style interpretation.

Together, these literatures motivate the paper’s three layers. Accounting and covenant design define what the engine must compute; discrimination evidence evaluates the ordering produced by the score; threshold and transport evidence evaluates the resulting rule in its intended setting. This paper’s narrower contribution is to use a synthetic IFRS 16 screening system to demonstrate why mechanical validation, discrimination and threshold portability should be assessed separately.

4 IFRS 16 covenant-screening setup

The outcome is broad financial failure over five annual observations. It is the union of nonpositive EBITDA, payment default, reserve funding deficit, leverage above 6.0, or interest coverage below 1.8. These are stipulated benchmark tests. Broad failure is consequently not interchangeable with payment default, and the covenant thresholds are not presented as the terms of observed loan contracts. Equality at a simulated covenant threshold is not a breach.

Let Lev_t^a and ICR_t^a denote analytic leverage and interest coverage at the annual testing dates. The minimum analytic headroom and fixed screening score are

$$H^a = \min \left\{ 6.0 - \max_t \text{Lev}_t^a, \min_t \text{ICR}_t^a - 1.8 \right\}, \quad z = \frac{1}{1 + \exp(2H^a)}. \quad (1)$$

Classification declares a positive when $z \geq \tau$. The comparator uses $\tau = 0.5$, including equality. The classification boundary and the strict simulated covenant-breach test therefore have different equality conventions.

Throughout the paper, z is called a screening score. It is not a calibrated probability of default. The analytic path and the financial simulation differ in timing, cash, financing and input conventions. Validation of the financial engine does not prove that the analytic path approximates the simulator within established error bounds. Nor can all cross-model differences be attributed to analytic approximation error alone. These distinctions constrain what can be inferred from agreement in rankings or disagreement in classifications.

Table 1: Stipulated synthetic archetype definitions

Template	Revenue	EBITDA	Debt	Lease	Cash
001	1200	260	520	390	45
002	980	205	470	360	35
003	1450	330	610	470	55
004	760	150	390	300	30
005	1680	390	650	500	60

Template	μ_g	σ_g	μ_m	Sweep	Lease run-off	Additions
001	3.0%	4.0%	22.0%	55.0%	11.0%	1.0%
002	2.5%	5.0%	21.0%	50.0%	12.0%	1.2%
003	3.5%	3.0%	23.0%	58.0%	10.0%	0.9%
004	2.0%	5.0%	20.0%	48.0%	13.0%	1.3%
005	4.0%	3.5%	24.0%	60.0%	10.0%	0.8%

All values are stipulated synthetic inputs; monetary quantities share the input file’s currency unit. Growth, margin, sweep and lease rates are parameters, not estimated sample frequencies.

5 Financial-engine mechanics and validation

The financial-engine checks provide a foundation for the subsequent screening evaluation. Independent geometric return and MOIC expectations match production within the recorded tolerance, and the independently derived debt schedule matches the production schedule. The checks address the reviewed implementation’s financial mechanics; they are not comparisons with external transaction outcomes. Table 2 summarizes the archived evidence.

All 1,000 benchmark scenario-years across 200 paths passed the recorded runtime invariants. The validation also detected six specified adversarial corruptions. The relevant invariants cover closing, cash, debt and roll-forward consistency on the tested paths. Deliberate corruption checks provide evidence that the validation detects the specified failures, rather than merely recording an uneventful execution.

This evidence has a precise scope. It supports tested accounting and mechanical consistency and independent return/debt sanity checks. It is not proof of correctness for every possible input, economic validity for every modeling convention, or validation against real borrower transactions. The engine section is therefore a prerequisite to interpreting the synthetic exercise, not the paper’s principal finding.

Table 2: Financial-engine validation evidence

Check	Archived evidence
Independent return	14.87% expected and actual
Independent MOIC	2.00 expected and actual
Debt schedule	Exact match to independent schedule
Runtime validation	200 paths; 1000 scenario-years
Recorded invariant failures	0
Adversarial corruption checks	6 specified corruptions detected

The recorded return comparison uses the archived tolerance. These checks establish consistency on tested paths, not correctness for every possible input or validation against external transactions.

6 Synthetic benchmark and evaluation protocol

The screening benchmark contains 200 scenarios across 5 synthetic templates and 20 broad financial failures. Table 1 reports the stipulated archetype inputs. The hotel-archetype labels identify synthetic structures, not observations of named borrowers. Parameter percentages in the input table describe stipulated growth, margin or financing conventions; they are not estimated subgroup event frequencies.

The score in Equation 1 remains fixed. In each leave-one-template-out fold, the threshold is selected using only the other templates and then applied to the excluded template. The objective is maximum training balanced accuracy, the average of sensitivity and specificity. The frozen tie rule selects the lowest threshold within the stated tolerance of the maximum. Test scores and labels do not participate in threshold selection.

The frozen candidates are the unique training scores, together with zero and the next representable value above one. The latter permits an all-negative classification even if a score equals one. The absolute tie tolerance is 10^{-12} . These rules specify a threshold-selection procedure; they do not fit or recalibrate the screening score. The comparison uses the original fixed threshold alongside the training-selected threshold on the same held-out rows.

Exclusion during threshold fitting is the relevant safeguard in this evaluation. It does not undo earlier choices in the construction of the synthetic benchmark, nor does it establish performance on new real operators. The experiment evaluates the transfer of the selected rule within the stipulated archetype system. It should not be described as validation of a learned borrower-default model.

7 Ranking performance versus threshold transfer

7.1 Strong discrimination of the fixed score

The screening score has aggregate ROC-AUC 0.947 and average precision 0.781 on the 200 scenarios, of which 20 fail. Figure 2 displays the ROC and precision-recall curves from the frozen scores. Average precision is the reported precision-recall summary; it is not trapezoidal area under that curve. The result supports strong ranking within this synthetic benchmark.

Aggregate held-out and pooled AUC are equal for a mechanical reason. The score is fixed and each original score-label pair appears once in the concatenated held-out rows. Excluding a template from threshold selection cannot change that aggregate ranking. The equality is therefore not an independent piece of evidence that discrimination transfers across templates. The new decision evidence is the behavior of the thresholds selected without the held-out template.

7.2 Recall improves at a material false-positive cost

Table 3 gives the aggregate confusion counts. Training-selected thresholds yield 15 true positives, 38 false positives, 142 true negatives and 5 false negatives. The fixed comparator yields 3 true

Fixed screening score: 200 scenarios, 20 failures

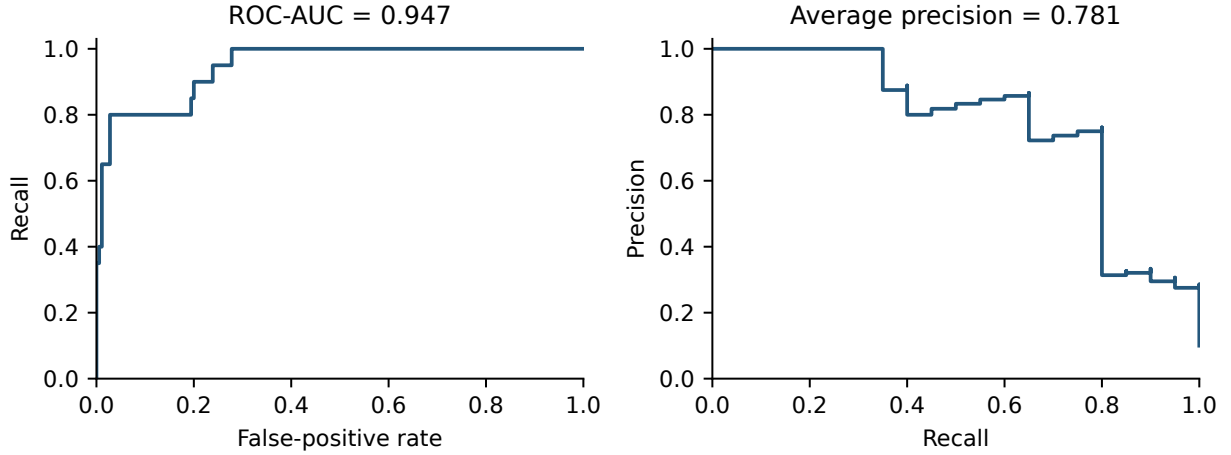


Figure 2: Ranking of the fixed screening score on 200 scenarios, including 20 failures and 180 non-failures. The curves use frozen scores with ties grouped; no fitted curve, smoothing or confidence interval is added. The precision–recall annotation is average precision. Equality with the pooled AUC follows from unchanged score–label pairs and is not independent validation.

positives, 0 false positives, 180 true negatives and 17 false negatives. Both comparisons concern the same 200 held-out scenarios.

Recall rises from 3/20 (15.0%) under the fixed comparator to 15/20 (75.0%) under transferred thresholds. Specificity falls from 180/180 (100.0%) to 142/180 (78.9%). Balanced accuracy rises from 0.575 to 0.769. The recall improvement must therefore be read together with the false-positive cost. Describing it simply as improved threshold performance would leave out a central part of the result.

Table 3: Aggregate threshold-transfer trade-off

Metric	Transferred threshold	Fixed threshold
True positives	15	3
False positives	38	0
True negatives	142	180
False negatives	5	17
Recall	15/20 (75.0%)	3/20 (15.0%)
Specificity	142/180 (78.9%)	180/180 (100.0%)
Balanced accuracy	0.769	0.575
Precision	15/53 (28.3%)	3/3 (100.0%)

Both policies classify the same 200 scenarios: 20 failures and 180 non-failures. The fixed threshold is 0.5. Rates show numerator/denominator; balanced accuracy averages recall and specificity. Aggregate ROC-AUC is 0.947 and average precision is 0.781 for the unchanged score under both policies. This identity is not independent generalization evidence.

The aggregate comparison does not establish a portable decision rule. It also does not establish economic superiority: that would require an explicit error-cost or utility model. The question addressed here is narrower—whether the training-only rule behaves consistently across the held-out structures. That question requires the template-level evidence, even though the aggregate balanced-accuracy objective improves.

8 Held-out template results

Table 4 and Figure 3 make the uneven transfer visible. Template 001 misses four of its five failures among 38 scenarios. Template 002 detects all three failures and falsely flags one of its 29 non-failures among 32 scenarios. Template 003 misses its only failure among 47 scenarios despite within-template AUC 1.000. A ranking estimate based on a single positive cannot support a precise inference about broader performance.

Template 004 contributes 37 of the 38 aggregate false positives. Its transferred threshold is 0.009678. Among its 49 scenarios, there are 11 failures and 38 non-failures; specificity is 2.6% (1/38) and the false-positive rate is 97.4% (37/38). Balanced accuracy is 0.513 under the transferred threshold, below 0.636 under the fixed comparator. This fold is the clearest example of aggregate improvement coexisting with poor local transfer.

Template 005 contains no failures among 34 scenarios. Its within-template AUC, average precision and balanced accuracy are undefined under the frozen policy, rather than evidence of perfect ranking. More generally, fold dispersion is descriptive. Training sets overlap and positive counts are sparse in some templates. The aggregate is not an equal-weight average of template metrics and cannot replace the examination of these counts.

Table 4: Held-out template ranking and threshold heterogeneity

Template	Train n/fail	Threshold	Train BA	Test n/fail	AUC	AP
001	162/15	0.047801	0.950	38/5	0.970	0.871
002	168/17	0.047352	0.869	32/3	0.989	0.917
003	153/19	0.047352	0.902	47/1	1.000	1.000
004	151/9	0.009678	0.954	49/11	0.978	0.934
005	166/20	0.047352	0.883	34/0	undef.	undef.

Template	TP	FP	TN	FN	Transfer BA	Fixed BA
001	1	0	33	4	0.600	0.500
002	3	1	28	0	0.983	0.500
003	0	0	46	1	0.500	0.500
004	11	37	1	0	0.513	0.636
005	0	0	34	0	undef.	undef.

BA: balanced accuracy; AP: average precision. Counts identify each class denominator (positives = TP+FN; negatives = TN+FP). Train/test cells give scenarios/failures. All threshold fitting excludes the test template. Template 003 contains one positive observation; AUC = 1.000 therefore means only that this single positive ranked above the 46 non-failures in the frozen realization and should not be interpreted as a precise estimate of discrimination. Undefined metrics for Template 005 are not zeros; this template has no failures and supports no within-template ranking inference. Thresholds are rounded for display only.

8.1 Cross-archetype heterogeneity

The secondary evidence is convergent and descriptive. Template 004 shows weak threshold transfer and comparatively weak financing-policy outcomes, providing evidence within the same synthetic system consistent with structural heterogeneity affecting downstream transfer. Its financing regime mix is 48/100 base, 31/100 downside and 21/100 distressed, close to the stipulated probabilities of 50%, 30% and 20%. Weakness persists after conditioning on regime, so the mixture alone does not explain it. Detailed cells appear in Appendix C.

This is not independent replication, external validation, causal evidence or empirical borrower evidence. The financing comparison remains a methodological demonstration, and regime-conditioned cells are small. The descriptive comparison does not show strict dominance on every risk metric: base and downside broad-failure rates under selected policies tie at zero. Template 004 has the lowest absolute opening cash and the highest debt/EBITDA among the stipulated

archetypes, but not the lowest cash/EBITDA. That distinction prevents an unsupported interpretation of the cash ratios.

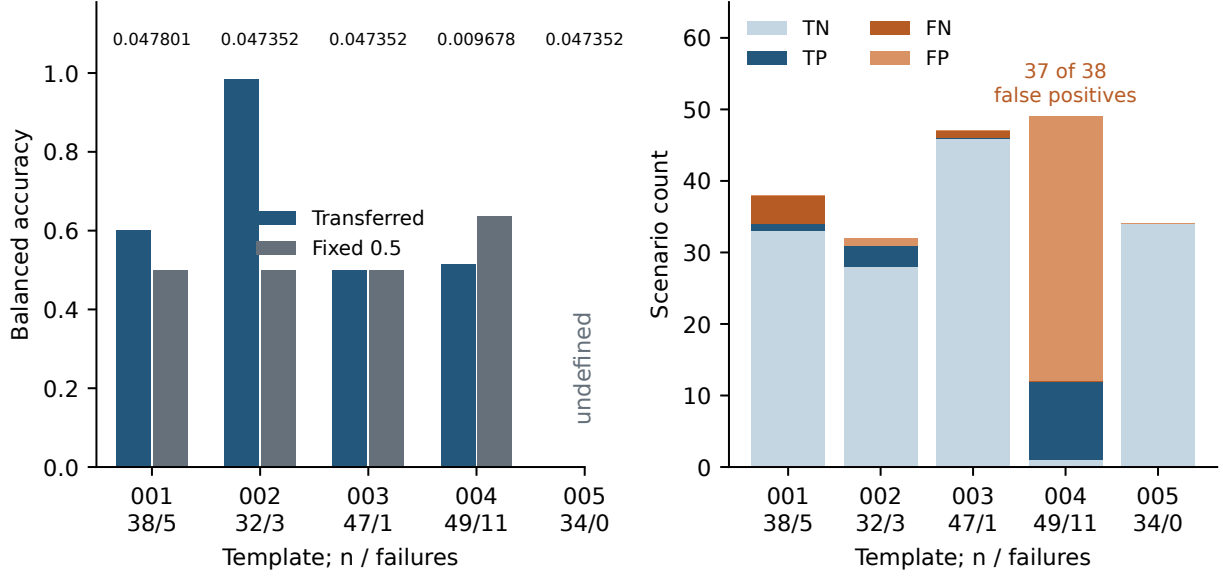


Figure 3: Threshold-transfer heterogeneity. Left: transferred and fixed-threshold balanced accuracy, with selected thresholds annotated above; Template 005 is undefined. Right: transferred-threshold confusion counts, including the concentration of 37/38 aggregate false positives in Template 004. Tick labels show total scenarios/failures; non-failure counts follow from their difference. Aggregated classification is not an equal-weight average across templates.

9 Limitations

The benchmark is a synthetic system with few archetypes and sparse failures in some folds. The overlapping training sets do not supply independent replications. The score is not a calibrated probability of default, and neither the ranking result nor the threshold comparison establishes economic utility. These limitations restrict the result to the tested setting; they are not qualifications that can be removed by reporting the aggregate AUC more precisely.

The uncertainty presentation is descriptive. No new confidence intervals, bootstrap results or significance tests are introduced. Subgroup rates are accompanied by their sample sizes or numerator/denominator, including the small regime-conditioned financing cells. Undefined metrics remain undefined. These reporting choices preserve the distinction between the observed realization and a claim about a population of borrower structures.

The heterogeneity synthesis has no causal identification and no independently replicated or external borrower evidence. The same archetype system appears in both decision procedures. Convergence within that system is informative descriptively, but it cannot identify which structural component causes the transfer difficulty or establish that the same mechanism operates in observed borrowers.

The financing-design experiment is retained as a methodological demonstration with stipulated assumptions. Its pervasive boundary selection and fixed-rate sensitivity restrict interpretation. There is no leverage-dependent credit pricing or recovery process, and the simulation continues mechanically after default. Entry and exit retain an asymmetric lease-treatment convention. These features preclude treating the selected policies as a substantive financing recommendation.

The Bayesian calibration experiment is excluded from the substantive results. Unverified calibration inputs and failed frozen technical and prior-sensitivity gates preclude posterior integration. This exclusion concerns the examined experiment and its evidence; it is not a general judgment that

Bayesian methods are invalid. Neither appendix changes the central screening result or supplies external borrower validation.

10 Conclusion

The synthetic benchmark separates three validation questions that should not be collapsed into one. The reviewed engine passes the archived mechanical checks on tested paths. The fixed screening score ranks adverse outcomes strongly. Yet that ranking does not establish a stable, portable threshold across structurally different synthetic archetypes. Passing one validation layer does not imply passing the next.

Training-only threshold selection improves aggregate recall at a false-positive cost, with pronounced template-level instability. The methodological implication is to evaluate decision thresholds explicitly under structural heterogeneity rather than infer their usefulness from discrimination alone. The conclusion is conditional on this synthetic system: it demonstrates neither successful deployment in real borrowers nor economic superiority of the selected rule.

A Financing-design experiment: methodological demonstration only

The financing-design experiment is retained as a methodological demonstration only. The reviewed screening benchmark is unsuitable for sponsor-return optimization because its entry valuation is revenue-based while exit valuation is EBITDA-based. The separate financing experiment uses EBITDA for both valuation bases. This consistency does not establish market calibration or remove the retained entry/exit lease-treatment limitation.

The separate experiment uses 100 operating scenarios per template and a frozen common policy grid. Each fold selects on the other four templates, using the same operating paths for candidates and the reference policy. Feasibility requires training broad-failure and payment-default rates no greater than the same-path reference rates. The objective is the training median limited-liability annualized sponsor return. Held-out results do not revise the grid or selection rule.

For raw engine exit equity E and positive initial sponsor equity S , the predeclared analysis-layer convention is

$$P = \max(0, E), \quad M = P/S, \quad r = M^{1/5} - 1. \quad (2)$$

Here P is sponsor exit proceeds, M is MOIC and r is annualized sponsor return. Zero recovery gives MOIC zero and $r = -1$, or -100% . Raw engine equity remains unfloored. This is a limited-liability sponsor-return convention, not an unconstrained financial IRR or a solver-failure convention.

Table A1 reports the methodological comparison. Across the same 500 held-out operating scenarios per policy, median annualized sponsor return is 1.10% under the selected policies and 0.61% under the reference policies. Broad failure is 1.4% (7/500) versus 6.8% (34/500), and payment default is 0.6% (3/500) versus 5.0% (25/500). All five folds select the 1.50 debt/EBITDA lower boundary; four of five select the 70% maximum sweep parameter.

Table A1: Synthetic financing-design methodological demonstration

Aggregate metric		Selected policy		Reference	
Median annualized return		1.10%		0.61%	
Mean annualized return		-1.11%		-3.26%	
Median MOIC		1.06		1.03	
Broad failure		7/500 (1.4%)		34/500 (6.8%)	
Payment default		3/500 (0.6%)		25/500 (5.0%)	
Total equity loss		1/500 (0.2%)		9/500 (1.8%)	

Template	Debt/EBITDA	Amort.	Sweep	Selected return	Ref. return
001	1.50	5.0%	70.0%	0.11%	-0.30%
002	1.50	7.5%	70.0%	0.86%	0.30%
003	1.50	5.0%	60.0%	1.82%	1.67%
004	1.50	7.5%	70.0%	-3.18%	-5.12%
005	1.50	5.0%	70.0%	2.74%	2.68%

Methodological demonstration only. Each policy is evaluated on the same 500 operating scenarios, with 100 per held-out template; the paired rows are not independent samples. Fold returns are medians (n=100 each); amortisation and sweep are stipulated policy parameters. Returns use the limited-liability sponsor convention. Grid-boundary selection is not an economic optimum.

Figure A1 shows a systematic within-grid preference for lower leverage across the feasible observed debt levels. It does not establish what happens below the lower boundary or at infeasible levels. Fixed-rate sensitivity changes policy components in 2/5 folds. The independent known-optimum toy, training-only checks and runtime invariants support search mechanics, but do not establish economic optimality. Missing endogenous credit pricing and recovery, together with mechanical continuation after default, keep the outcome methodological.

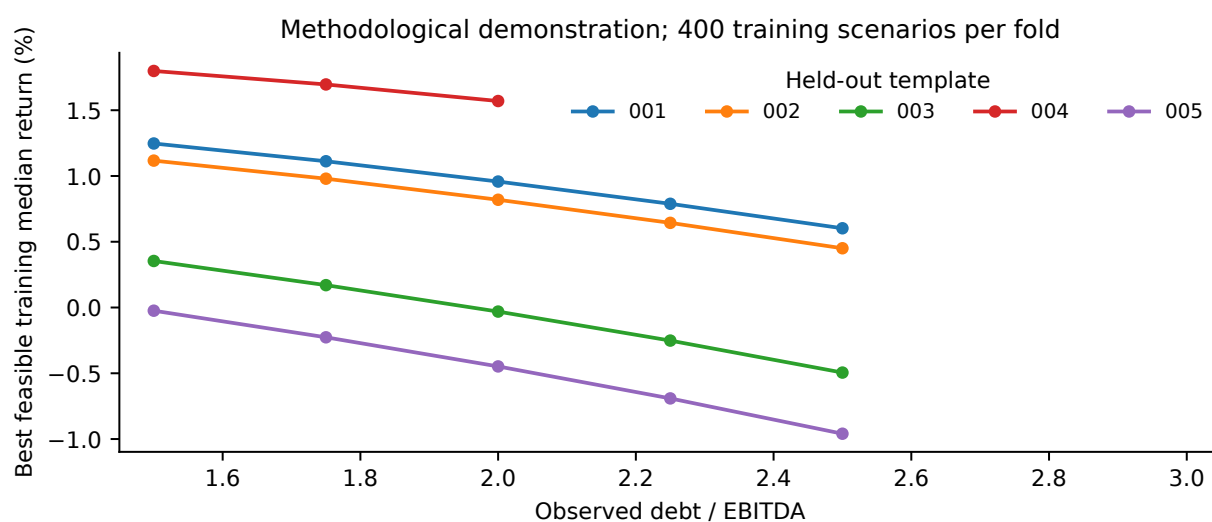


Figure A1: Feasible observed debt-objective profile in the methodological financing demonstration. Each fold uses 400 training scenarios. Points are archived best-feasible objectives; infeasible levels are missing, and no curve extends below 1.50. Connecting observed feasible levels is a display aid, not an estimated surface. The within-grid pattern does not establish an economic optimum.

B Excluded Bayesian calibration experiment

The Bayesian calibration experiment is excluded from the substantive results. The corrected experiment examined synthetic recovery, but the ten named-firm calibration rows remained unverified/stipulated. Firm names are not variable-level provenance. The recorded convergence/effective-sampling and prior-sensitivity gates did not support integration, and no posterior samples were used in the LBO results.

Table A2 places the failed gates alongside the diagnostic evidence. Attractive synthetic recovery does not overcome poor provenance and failed admission criteria. The relevant methodological conclusion is exclusion of this experiment from substantive results, not an empirical calibration finding. A recovery exercise with stipulated inputs cannot supply the missing evidentiary basis for those inputs.

This decision does not imply that Bayesian population modeling is intrinsically invalid. It bounds what the current evidence permits. The exclusion is preserved here because it limits the admissible interpretation of the broader workflow, not because the excluded method is a principal contribution of the paper.

Table A2: Bayesian exclusion diagnostics

Diagnostic	Archived result
Recovery runs	18
Parameter evaluations	180
Maximum R-hat	1.015
Minimum bulk ESS	391.486
Minimum tail ESS	176.280
Divergences	0
Frozen R-hat gate	Failed
Frozen tail-ESS gate	Failed
Material prior sensitivity	Present
Input provenance	Unverified/stipulated
Substantive-results treatment	Excluded

The 18 recovery runs yield 180 parameter evaluations, not 180 independent datasets. Calibration and prior sensitivity use ten stipulated named-firm rows. Synthetic recovery does not override the failed diagnostic and prior-sensitivity gates; posterior samples were not integrated into LBO results.

C Regime diagnostics and reproducibility

C.1 Small-cell financing diagnostics

Table A3 reports the Template 004 regime cells and the selected-policy median-return comparison across templates. The numbers are descriptive outcomes from the same synthetic system. The cell counts appear beside all return percentages, and event rates show counts and denominators. Figure A2 presents the within-regime comparison without confidence bars. Template 004’s selected-policy median return is the lowest among the templates in each regime, while some risk metrics tie.

In the distressed cell, $n = 21$. Selected policies yield broad failures 7/21 (33.3%), payment defaults 3/21 (14.3%), insolvencies 6/21 (28.6%), covenant breaches 7/21 (33.3%), and total equity loss 1/21 (4.8%). Reference policies yield the respective counts 18/21 (85.7%), 16/21 (76.2%), 18/21 (85.7%), 14/21 (66.7%), and 7/21 (33.3%). The small denominator is integral to the interpretation of each percentage.

Template 004 reference distressed P10 annualized sponsor return is -100% ($n = 21$). Seven of the 21 scenarios have zero recovery, so the 10th percentile lies on that zero-recovery mass. This is mechanically consistent with the predeclared limited-liability convention. It does not mean raw equity is bounded at -100% , that an unconstrained IRR was calculated, or that a solver failed.

Table A3: Template 004 regime-conditioned outcomes and within-regime comparison

	Template 004 cell	n	Median return	P10 return
	Selected base	48	2.58%	-3.71%
	Selected downside	31	-7.84%	-16.58%
	Selected distressed	21	-18.85%	-36.80%
	Reference base	48	2.16%	-5.81%
	Reference downside	31	-11.44%	-25.48%
	Reference distressed	21	-30.00%	-100.00%

Cell	n	Broad fail	Pay default	Insolvency	Covenant	Equity loss
Selected base	48	0/48 (0.0%)	0/48 (0.0%)	0/48 (0.0%)	0/48 (0.0%)	0/48 (0.0%)
Selected downside	31	0/31 (0.0%)	0/31 (0.0%)	0/31 (0.0%)	0/31 (0.0%)	0/31 (0.0%)
Selected distressed	21	7/21 (33.3%)	3/21 (14.3%)	6/21 (28.6%)	7/21 (33.3%)	1/21 (4.8%)
Reference base	48	1/48 (2.1%)	1/48 (2.1%)	1/48 (2.1%)	1/48 (2.1%)	0/48 (0.0%)
Reference downside	31	9/31 (29.0%)	7/31 (22.6%)	9/31 (29.0%)	3/31 (9.7%)	1/31 (3.2%)
Reference distressed	21	18/21 (85.7%)	16/21 (76.2%)	18/21 (85.7%)	14/21 (66.7%)	7/21 (33.3%)

Selected-policy template	Base return (n)	Downside return (n)	Distressed return (n)
001	7.84% (41)	-4.24% (41)	-16.03% (18)
002	3.76% (62)	-5.09% (24)	-15.29% (14)
003	7.68% (55)	-1.79% (29)	-14.00% (16)
004	2.58% (48)	-7.84% (31)	-18.85% (21)
005	8.71% (42)	1.34% (42)	-10.44% (16)

Methodological financing demonstration in the same synthetic system; cells are descriptive and small. Risk entries are count/n (percent); return percentages have n in the same row. Template 004 regime counts are 48/100, 31/100 and 21/100, close to stipulated probabilities 50/30/20 percent. Reference distressed P10 is the limited-liability floor: 7/21 zero recoveries (33.3 percent) place it at -100 percent. No confidence intervals are inferred.

C.2 Screening score distributions

Figure A3 displays the frozen screening scores directly by template and outcome. No density smoothing or score recalibration is used. The transferred threshold is shown for each template, with total and positive counts retained. The display supports the ranking-versus-threshold distinction without treating a score value as an estimated default probability.

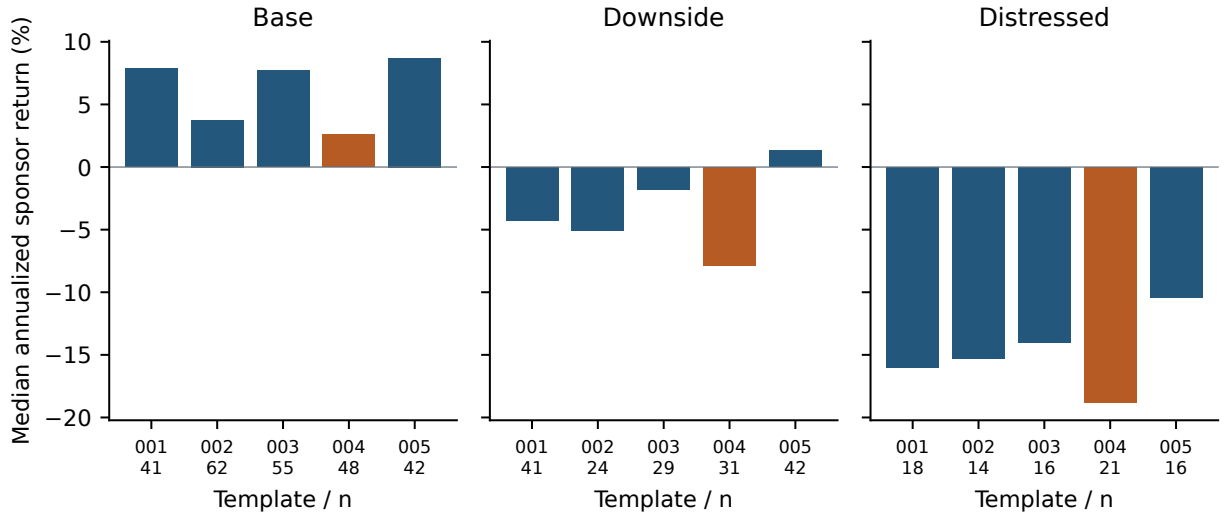


Figure A2: Selected-policy median annualized sponsor return within each realized regime; tick labels show template and n. Template 004 is highlighted. The financing experiment remains methodological, and the comparison is descriptive within the same synthetic system, not independent replication or causal evidence. Distressed Template 004 has n=21. No uncertainty bars are inferred.

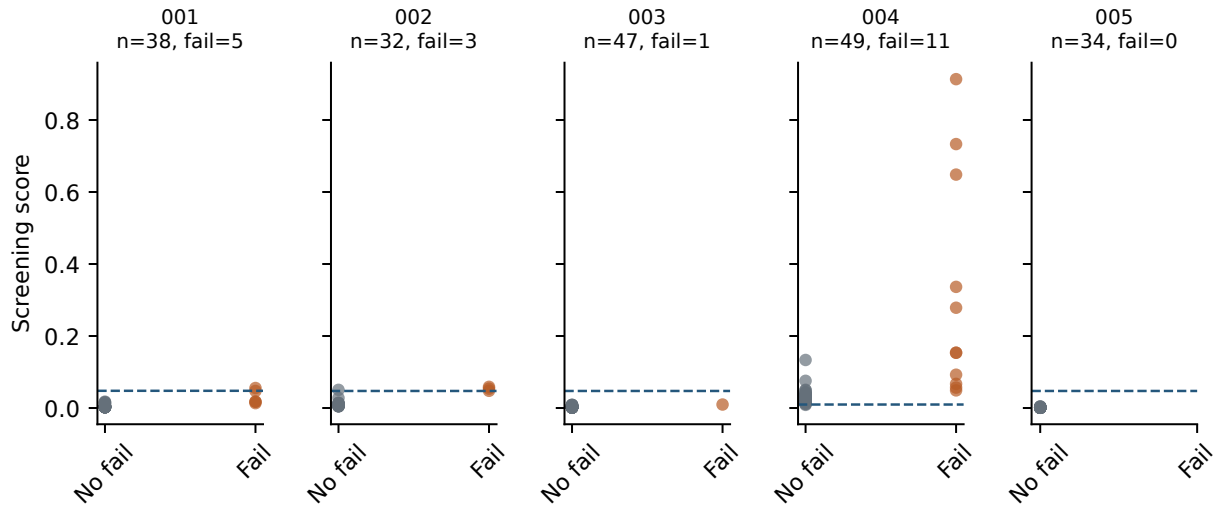


Figure A3: Raw screening scores by template and failure status. Titles give n and failure counts; non-failure counts are the difference. Dashed lines mark transferred thresholds. Overlapping points retain their original values; no smoothing or uncertainty intervals are added. The single positive in Template 003 and absence of positives in Template 005 restrict within-template ranking inference.

C.3 Artifact lineage and rendering

The archived baseline reproduced its financial and statistical fields exactly, excluding runtime timing and source-commit metadata. Repository reproducibility does not establish external validity. The present manuscript uses frozen numerical outputs and adds only rendering, exposition and build verification.

The claim-freeze commit is 21d726606f3208bcee6e3e84ae4c357e3a33d79b; the manuscript source commit is d3501d9fd0c944d9fcd631216f48d4a511c5e821. The corrected pre-freeze source is 0f521a28150fe56f98cdabb3d1cb3664286df124. Input and artifact hashes are preserved in the freeze source manifest, with protocol lineage in the freeze summary. Synthetic input definitions reside in `data/synthetic/operators.csv`. The screening benchmark uses seed 42; the separate financing demonstration uses seed 314159. These identify archived realizations and are not new simulation runs.

The rendering and PDF build commands are:

```
python -m analysis.scripts.render_manuscript_v3
python -m analysis.scripts.build_manuscript_v3
```

The renderer reads the frozen specifications and source artifacts. The build uses Tectonic, the toolchain documented for the prior manuscript. Rendering records individual numeric fields and transformations; source comments retain claim links without exposing claim identifiers in the PDF. The final build audit records source and PDF hashes, exhibit counts, citation/reference checks and protected-artifact verification.

C.4 Code and data availability

Code, synthetic inputs, frozen validation artifacts and reproduction instructions are available in the accompanying public repository at <https://github.com/Aniket2002/ifrs16-lbo-engine>. The repository branch `v3/validated-rebuild` contains the manuscript lineage; the exact manuscript-source commit is recorded above and in the build audit. No archival DOI is claimed.

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